

上节回顾

良导体 $\left\{ \begin{array}{l} \text{GHz以下频段} \quad S = \sqrt{\frac{2}{\sigma \omega \mu_0}} \quad \text{当 } \delta \downarrow \\ \text{微波段} \quad \omega_B \gg \omega_E \\ \omega < \omega_p \text{ 时 电磁波不能传播} \quad \delta = \sqrt{\frac{c^2}{\omega_p^2 - \omega^2}} \\ \omega = \omega_p \text{ 时} \quad \delta \rightarrow \infty \\ \omega > \omega_p \text{ 时 电磁波可以传播} \quad 0 < \epsilon_r < 1 \text{ 比真空更光滑} \end{array} \right.$

非良导体 $\left(\frac{\sigma_c}{\epsilon_0 \omega} \ll 1 \right)$ 穿透深度 $\delta = \frac{2}{\sigma_c} \sqrt{\frac{\epsilon}{\mu_0}}$ (与 ω 无关)

波导中的场方程和边界条件

理想导体界面的边界条件: $\vec{B}$ 沿切向, $\vec{E}$ 沿法向

波导中的场方程 $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + k_c^2 \right) \left(\frac{\vec{E}_0}{B_0} \right) = 0$

E_x, E_y, B_x, B_y 可由 E_z, B_z 确定

E_z, B_z 不能同时为 0. 即波导中的电磁波不是横电磁波 (TEM)

1. $B_z \neq 0, E_z = 0$: 横电波 (TE 波)

2. $B_z = 0, E_z \neq 0$: 横磁波 (TM 波)

TE 波: 截止频率 ω_c . 只有 $\omega > \omega_c$ 的波才可以在波导中传播.

$$\omega_c = ck_c = c\pi \sqrt{\frac{m^2}{a^2} + \frac{n^2}{b^2}}$$

截止波长 λ_c . 只有 $\lambda < \lambda_c$ 的波

$$\lambda_c = \frac{2\pi}{k_c} = \frac{2}{\sqrt{\frac{m^2}{a^2} + \frac{n^2}{b^2}}} = \frac{2ab}{\sqrt{m^2 b^2 + n^2 a^2}}$$

例. 频率为 $30 \times 10^9 \text{ Hz}$ 的微波, 在 $0.7 \text{ cm} \times 0.4 \text{ cm}$ 的矩形波导中能以什么波模传播?

$\omega = 2\pi f = 2\pi \times 30 \times 10^9 \text{ Hz} \quad \lambda = \frac{2\pi c}{\omega} = \frac{2\pi c}{2\pi \times 3 \times 10^{10}} = \frac{c}{3 \times 10^{10}} = 1 \text{ cm}$

TE₁₀ 模: $m=1, n=0 \quad \lambda_c = \frac{2 \times 0.7 \times 0.4}{0.4} = 1.4 \text{ cm} > 1 \text{ cm}$ 可传播

TE₀₁ 模: $m=0, n=1 \quad \lambda_c = \frac{2 \times 0.7 \times 0.4}{0.7} = 0.8 \text{ cm} < 1 \text{ cm}$ 不可传播

TM 模的最低阶为 TM₁₁. 不可传播

对于 $0.7 \text{ cm} \times 0.6 \text{ cm}$ 的波导管呢? TE₁₀ 和 TE₀₁ 模.

能流. H_z 与其它分量间存在 $\pi/2$ 相位.

$$\vec{S} = \vec{E} \times \vec{H} \quad \text{与 } \langle S_z \rangle \neq 0$$

作业: 求解 TM 波.

谐振腔.

谐振腔是由理想导体围成的封闭腔体.

波导 + 端面.

~~波导中~~ TE 波 $B_z = B_0 \cos(\frac{m\pi}{a}x) \cos(\frac{n\pi}{b}y) e^{i(k_z z - \omega t)}$

端面对电磁波进行反射.

$$B_z = [B_0 \cos(\frac{m\pi}{a}x) \cos(\frac{n\pi}{b}y) e^{ik_z z} + B'_0 \cos(\frac{m\pi}{a}x) \cos(\frac{n\pi}{b}y) e^{-ik_z z}] e^{-i\omega t}$$

边界条件 $B_z|_{z=0, d} = 0$

$$z=0 \text{ 时 } B_z = 0 \Rightarrow B_0 + B'_0 = 0$$

$$z=d \text{ 时 } B_z = 0 \Rightarrow \sin(k_z d) = 0 \Rightarrow k_z d = p\pi$$

$$\begin{cases} B_x = -2iB_0 \frac{m\pi}{a} \frac{p\pi}{d} \frac{1}{k_c^2} \sin(\frac{m\pi}{a}x) \cos(\frac{n\pi}{b}y) \cos(\frac{p\pi}{d}z) e^{-i\omega t} \\ B_y = -2iB_0 \frac{n\pi}{b} \frac{p\pi}{d} \frac{1}{k_c^2} \cos(\frac{m\pi}{a}x) \sin(\frac{n\pi}{b}y) \cos(\frac{p\pi}{d}z) e^{-i\omega t} \\ B_z = 2iB_0 \cos(\frac{m\pi}{a}x) \cos(\frac{n\pi}{b}y) \sin(\frac{p\pi}{d}z) e^{-i\omega t} \\ E_x = 2B_0 \frac{n\pi}{b} \frac{\omega}{k_c^2} \cos(\frac{m\pi}{a}x) \sin(\frac{n\pi}{b}y) \sin(\frac{p\pi}{d}z) e^{-i\omega t} \\ E_y = -2B_0 \frac{m\pi}{a} \frac{\omega}{k_c^2} \sin(\frac{m\pi}{a}x) \cos(\frac{n\pi}{b}y) \sin(\frac{p\pi}{d}z) e^{-i\omega t} \\ E_z = 0 \end{cases}$$

讨论: 1) 电场和磁场相差相位 $\frac{\pi}{2}$ $\langle S \rangle = 0$

$$2) k^2 = k_c^2 + k_z^2 = (\frac{m\pi}{a})^2 + (\frac{n\pi}{b})^2 + (\frac{p\pi}{d})^2$$

$$\omega = kc = c\pi \sqrt{\frac{m^2}{a^2} + \frac{n^2}{b^2} + \frac{p^2}{d^2}} \quad \lambda = \frac{2\pi}{k} = \frac{2}{\sqrt{\frac{m^2}{a^2} + \frac{n^2}{b^2} + \frac{p^2}{d^2}}}$$

3) $p \neq 0$. m, n, p 中最多 1 个为 0. 谐振腔的最低阶模式为 (110) 或 (101) 或 (011)

4) 一个立方腔 (边长为 a) 的谐振腔, 最低阶的腔膜波长为 $\lambda = \sqrt{2}a$.

要实现某一波长的谐振功能, 其尺度必须大于某一特征大小.

电磁波的辐射

①

1. 电磁场的矢势和标势

真空:
$$\begin{cases} \nabla \cdot \vec{E} = \rho/\epsilon_0 \\ \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \\ \nabla \cdot \vec{B} = 0 \\ \nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} = \mu_0 \vec{J} + \frac{1}{c^2} \frac{\partial \vec{E}}{\partial t} \end{cases}$$

位移电流 $\frac{\partial \vec{E}}{\partial t}$ 使电磁辐射成为可能

$$\nabla \cdot \vec{B} = 0 \Rightarrow \vec{B} = \nabla \times \vec{A}$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \Rightarrow \nabla \times \vec{E} = -\nabla \times \frac{\partial \vec{A}}{\partial t}$$

$$\nabla \times (\vec{E} + \frac{\partial \vec{A}}{\partial t}) = 0$$

$$\vec{E} + \frac{\partial \vec{A}}{\partial t} = -\nabla \varphi$$

$$\vec{E} = -\nabla \varphi - \frac{\partial \vec{A}}{\partial t}$$

$\vec{A}$ 与 φ 不唯一

$$\vec{A}' = \vec{A} + \nabla \varphi \Rightarrow \vec{B} = \nabla \times (\vec{A} + \nabla \varphi) = \nabla \times \vec{A}$$

$$\begin{aligned} \varphi' &= \varphi - \frac{\partial \varphi}{\partial t} \Rightarrow \vec{E} = -\nabla \varphi' - \frac{\partial \vec{A}'}{\partial t} = -\nabla (\varphi - \frac{\partial \varphi}{\partial t}) - \frac{\partial \vec{A}}{\partial t} - \frac{\partial \nabla \varphi}{\partial t} \\ &= -\nabla \varphi - \frac{\partial \vec{A}}{\partial t} \end{aligned}$$

即 $(\vec{A}, \varphi)$ 与 $(\vec{A}', \varphi')$ 对应同样的 $\vec{B}, \vec{E}$.

为了消除这种任意性, 施加一限制 (规范条件)

1) 库伦规范 $\nabla \cdot \vec{A} = 0$

2) 洛伦兹规范 $\nabla \cdot \vec{A} + \frac{1}{c} \frac{\partial \varphi}{\partial t} = 0$

2. 达朗贝尔方程

由麦克斯韦方程组导出 $\vec{A}$ 和 φ 满足的方程

$$\nabla \times \vec{B} = \nabla \times (\nabla \times \vec{A}) = \mu_0 \vec{J} + \frac{1}{c^2} \frac{\partial \vec{E}}{\partial t} = \mu_0 \vec{J} + \frac{1}{c^2} \frac{\partial}{\partial t} (-\nabla \varphi - \frac{\partial \vec{A}}{\partial t})$$

$$\nabla (\nabla \cdot \vec{A}) - \nabla^2 \vec{A} = \mu_0 \vec{J} + \frac{1}{c^2} \frac{\partial}{\partial t} (\nabla \varphi + \frac{\partial \vec{A}}{\partial t}) \Rightarrow \nabla (\nabla \cdot \vec{A} + \frac{1}{c} \frac{\partial \varphi}{\partial t}) - \nabla^2 \vec{A} + \frac{1}{c^2} \frac{\partial^2 \vec{A}}{\partial t^2}$$

$$\nabla \cdot \vec{E} = \nabla \cdot (-\nabla \varphi - \frac{\partial \vec{A}}{\partial t}) = -\nabla^2 \varphi - \frac{\partial}{\partial t} \nabla \cdot \vec{A} = \rho/\epsilon_0$$

若采用洛伦兹规范, 则

$$\begin{aligned}\nabla^2 \vec{A} - \frac{1}{c^2} \frac{\partial^2 \vec{A}}{\partial t^2} &= -\mu_0 \vec{J} \\ \nabla^2 \varphi - \frac{1}{c^2} \frac{\partial^2 \varphi}{\partial t^2} &= -\rho/\epsilon\end{aligned}\quad (\nabla \cdot \vec{A} + \frac{1}{c} \frac{\partial \varphi}{\partial t} = 0)$$

$\vec{A}$ 与 φ 满足相同形式的方程, 达朗贝尔方程, 非齐次波动方程.

$\rho \rightarrow \varphi, \vec{J} \rightarrow \vec{A}$. 离开 $\rho, \vec{J}$, φ 与 $\vec{A}$ 以波动形式在空间传播.

3. 推迟势

$$\text{求解 } \nabla^2 \varphi - \frac{1}{c^2} \frac{\partial^2 \varphi}{\partial t^2} = -\rho/\epsilon$$

思路: 点电荷 + 叠加原理

$$\nabla^2 \varphi - \frac{1}{c^2} \frac{\partial^2 \varphi}{\partial t^2} = -\frac{Q(t) \delta(\vec{r})}{\epsilon} \quad [\text{在 } \vec{r}=0 \text{ 处有一正电荷 } Q(t)]$$

由球对称性, φ 只依赖于 r, t .

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \varphi}{\partial r} \right) - \frac{1}{c^2} \frac{\partial^2 \varphi}{\partial t^2} = -\frac{1}{\epsilon} Q(t) \delta(r)$$

在 $r \neq 0$ 处

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \varphi}{\partial r} \right) - \frac{1}{c^2} \frac{\partial^2 \varphi}{\partial t^2} = 0$$

$$\text{令 } \varphi(r, t) = \frac{u(r, t)}{r}, \text{ 则 } \frac{\partial \varphi}{\partial r} = \frac{\partial u}{\partial r} \cdot \frac{1}{r} - \frac{u}{r^2}$$

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left[r^2 \left(\frac{1}{r} \frac{\partial u}{\partial r} - \frac{u}{r^2} \right) \right] - \frac{1}{rc^2} \frac{\partial^2 u}{\partial t^2} = 0$$

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left[r \frac{\partial u}{\partial r} - u \right] - \frac{1}{rc^2} \frac{\partial^2 u}{\partial t^2} = 0$$

$$\frac{1}{r^2} \left[\cancel{\frac{\partial u}{\partial r}} + r \frac{\partial^2 u}{\partial r^2} - \cancel{\frac{\partial u}{\partial r}} \right] - \frac{1}{rc^2} \frac{\partial^2 u}{\partial t^2} = 0$$

$$\frac{\partial^2 u}{\partial r^2} - \frac{1}{c^2} \frac{\partial^2 u}{\partial t^2} = 0 \quad (\text{一维空间中的波动方程})$$

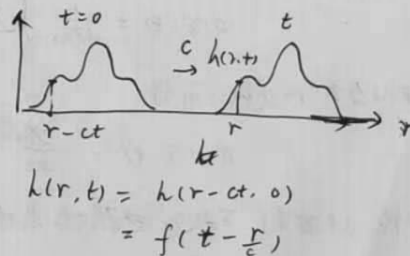
$$\text{通解为 } u(r, t) = f\left(t - \frac{r}{c}\right) + g\left(t + \frac{r}{c}\right)$$

波: 均匀介质中的一个扰动以同区的形状和一定的速度传播.

$$\text{可以证明 } f\left(t - \frac{r}{c}\right) \text{ 满足 } \frac{\partial^2 f}{\partial r^2} - \frac{1}{c^2} \frac{\partial^2 f}{\partial t^2} = 0$$

证明: 设 $\alpha = t - \frac{r}{c}$.

$$\frac{\partial f}{\partial r} = \frac{\partial f}{\partial \alpha} \frac{\partial \alpha}{\partial r} = \frac{\partial f}{\partial \alpha} \left(-\frac{1}{c}\right), \quad \frac{\partial^2 f}{\partial r^2} = \left(-\frac{1}{c}\right)^2 \frac{\partial^2 f}{\partial \alpha^2}, \quad \frac{\partial f}{\partial t} = \frac{\partial f}{\partial \alpha} \frac{\partial \alpha}{\partial t} = \frac{\partial f}{\partial \alpha}, \quad \frac{\partial^2 f}{\partial t^2} - \frac{1}{c^2} \frac{\partial^2 f}{\partial r^2} = 0$$



③

$$\varphi(\vec{r}, t) = \frac{f(t - \frac{r}{c})}{r} + \frac{g(t + \frac{r}{c})}{r} \quad (\vec{r} \neq 0)$$

从原点上向外发射 向原点上会聚

考虑辐射问题

$$\varphi(\vec{r}, t) = \frac{f(t - \frac{r}{c})}{r}$$

下面由包含 $t=0$ 的方程确定 f 的函数形式

把 $\varphi(\vec{r}, t) = f(t - \frac{r}{c})/r$ 代入 $\nabla^2 \varphi - \frac{1}{c^2} \frac{\partial^2 \varphi}{\partial t^2} = -\frac{1}{\epsilon_0} \rho(\vec{r}, t)$

以球心为坐标原点, 半径为 $\epsilon \rightarrow 0$ 的球体 V_ϵ 内作积分.

$$\int_{V_\epsilon} \left[\nabla^2 \frac{f}{r} - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \frac{f}{r} \right] dV = -\frac{1}{\epsilon_0} \int_{V_\epsilon} \rho(t) \delta(\vec{r}) dV$$

$$\nabla^2 \frac{f}{r} = \nabla \cdot \left(\frac{\nabla f}{r} + f \nabla \frac{1}{r} \right) = \frac{1}{r} \nabla^2 f + 2 \nabla f \cdot \nabla \frac{1}{r} + f \nabla^2 \frac{1}{r}$$

利用 $\nabla^2 \frac{1}{r} = -4\pi \delta(\vec{r})$, 得

$$\int_{V_\epsilon} \left[\frac{1}{r} \nabla^2 f + 2 \nabla f \cdot \nabla \frac{1}{r} - 4\pi f \delta(\vec{r}) - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \frac{f}{r} \right] dV = -\frac{Q(t)}{\epsilon_0}$$

$\epsilon \rightarrow 0$

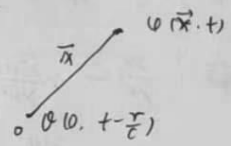
$$-4\pi f(t) = -\frac{Q(t)}{\epsilon_0}$$

$$f(t) = \frac{Q(t)}{4\pi \epsilon_0}$$

$$\varphi(\vec{r}, t) = \frac{f(t - \frac{r}{c})}{r} = \frac{Q(t - \frac{r}{c})}{4\pi \epsilon_0 r} = \frac{Q(\vec{r}=0, t - \frac{r}{c})}{4\pi \epsilon_0 r}$$

位于 $\vec{r}$ 处的电荷激发的势

$$\varphi(\vec{r}, t) = \frac{Q(\vec{r}, t - \frac{r}{c})}{4\pi \epsilon_0 r}$$



任意电荷分布激发的势

$$\varphi(\vec{r}, t) = \frac{1}{4\pi \epsilon_0} \int \frac{\rho(\vec{r}', t - \frac{r}{c})}{r} dV'$$

推迟势

由 φ 与 A 的类比, 可得

$$A(\vec{r}, t) = \frac{\mu_0}{4\pi} \int \frac{\vec{J}(\vec{r}', t - \frac{r}{c})}{r} dV'$$

讨论: 1) t 时刻, $\vec{r}$ 处的电磁场是由较早时刻 $t' = t - \frac{r}{c}$ 处的电荷和电流分布决定的

2) 推迟的时间 $t - t' = \frac{r}{c}$, 可见电磁作用是以光速传递的

辐射场的计算

$$\vec{J} \Rightarrow \vec{A} \Rightarrow \vec{B} = \nabla \times \vec{A} \quad \nabla \cdot \vec{J} = -\frac{\partial \rho}{\partial t} = i\omega \rho \Rightarrow \vec{E} \quad \text{在 } \vec{J}=0 \text{ 的区域} \\ \nabla \times \vec{B} = \frac{1}{c^2} \frac{\partial \vec{E}}{\partial t} = -i\frac{\omega}{c^2} \vec{E} \Rightarrow \vec{E} = i\frac{1}{k} \nabla \times \vec{B}$$